Detect Ransomware Before It Encrypts Files by Analyzing System Behavior Using Deep Learning Models

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***Abstract*—In the present world, ransomware is considered to be the most dangerous type of malware that has caused losses amounting to billions of dollars due to file encryption and ransom demand from victims. Signature based detection mechanisms are ineffective in detecting new polymorphic ransomware threats prior to their harmful activities. The proposed methodology in this paper involves the use of deep learning models for proactive ransomware detection through analysis of sequences of API calls generated by processes. Three models – Hybrid CNN-LSTM model, Multi-kernel CNN 3-scales model and Multi-kernel CNN 5-scales model have been used and compared against each other using a balanced dataset consisting of 85,594 API calls sequence data. The best model performs with 99.59 percent detection accuracy. In addition, an Early Detection Analysis has been carried out which measures the efficiency of the models at different observation intervals. It was found that the Multi-kernel CNN 5-scales model is able to detect ransomware accurately 77.8 percent of the times after observing just 20 API calls, much before the encryption process takes place**

***Index Terms*—ransomware detection, deep learning, LSTM, CNN, API call Sequences, endpoint security, behavioral analysis, Early detection, Multi-Kernel CNN**

I. Introduction

Ransomware can be defined as a malicious software that locks user's files and demands a ransom in the form of cryptocurrency for decryption. In the last ten years, ransomware has evolved from simple software targeting consumers to highly complex campaigns that target hospitals, government agencies, and critical infrastructure. Financial damage caused by ransomware surpassed 30 dollar billion in 2023 and keeps rising with the development of Ransomware-as-a-Service services which enable unskilled attackers to run extremely damaging attacks.

Currently, detection approaches primarily rely on two methods of ransomware detection include signature-based detection and backup & recovery strategy. Signature-based detection utilizes antivirus software that matches file hashes or byte sequences with databases of known malwares. This method effectively identifies previously recognized malware and fails when facing unknown (zero-day or polymorphic) versions of ransomware which alter themselves in order to bypass antivirus programs. Backup recovery is an approach that is focused on mitigating damage rather than preventing an attack.

However, behavioral detection appears to be a more viable solution. Regardless of its specific implementation, ransomware should perform a predictable set of actions: enumerate directories, read files, call crypto functions and write encrypted data to a hard drive.This behavioral fingerprint remains consistent even when the underlying code changes, making it a reliable detection target.

Deep learning models, particularly Long Short-Term Memory networks and Convolutional Neural Networks, have demonstrated strong capability in learning patterns from sequential data. In this work, we propose a deep learning framework that analyzes Windows API call sequences in real time to classify running processes as ransomware or benign before the encryption phase begins. We compare three architectures — CNN-LSTM, 3-scale multi-kernel CNN, and 5-scale multi-kernel CNN — and introduce an Early Detection Analysis to quantify how quickly our models can raise an alert after process execution begins.

II. Related Works

Several research papers based on behavioral and deep learning models for detecting ransomware and malware attacks have been created lately. A systematic survey on ransomware detection via deep learning from 2018 to 2024 has been conducted in [1]. As a result, it has been shown that hybrid deep learning algorithms work better than single algorithms and dynamic behavioural analysis works better than static analysis in the detection of novel ransomware. Paper [2] examined the potential of CNN and LSTM models for detection of ransomware attacks via analysis of API call sequences. Different window sizes have been tested for experiments; it has been found out that using the window size of 7 API calls leads to better performance of deep learning models which outperform traditional machine learning algorithms. In paper [3], the efficiency of LSTM and BERT models for detecting the behavior of ransomware attacks has been tested via analysis of API call sequences. It emphasizes the capability of transformer and recurrent neural network models for correctly recognizing the order of API calls for ransomware with high accuracy rates. XRan is the explainable deep learning method for detection of ransomware attacks based on dynamic behavior analysis.

The model includes API call sequence analysis and XAI-based decisions about ransomware attacks [4].

The authors of [5] have suggested the usage of a multi-head self-attention long short-term memory network to detect ransomware in the early stages. It was shown that the attention mechanism can greatly increase the performance of the model since it is able to find the most discriminant API calls in a sequence. There has been conducted a comprehensive survey on ransomware detection techniques and review of related papers from 2017 to 2022, along with the discussion of current tendencies and future perspectives in ransomware detection in [6]. A systematic evaluation framework for the assessment of machine learning and deep learning based ransomware detection methods has been suggested in [7] by the authors of this paper, as well as identified research methodology gaps and future research directions. The authors of [8] have used deep learning for classification of malware system call sequences using hybrid CNN-RNN architecture.

The research in [9] brought about the concept of data augmentation for malware detection by means of analysis of API call sequence using convolutional neural network, showing improved performance by creating artificial data. The data-centric (victim-centric) ransomware detection system named CryptoDrop, which analyzes filesystem activities using such factors as file type changes, Shannon entropy, and file similarity, was suggested in [10] with the idea to detect ransomware with minimum damage to the victim's computer. The research in [11] resulted in introducing UNVEIL, the automated ransomware detection system by behavior analysis of filesystems and desktop interaction in a sandboxed environment. The LSTM model in [12] demonstrated the possibility of using recurrent neural networks to capture the long-term dependencies in sequential data and laid out theoretical foundations for the possible usage of sequential modeling in machine learning tasks, among other things, for analyzing API call sequences for malware detection. DeepLog, the system proposed in [13], based on deep learning approach for anomaly detection and diagnostics using system logs, proves the efficiency of using LSTM networks for sequential analysis of system events.

The usage of deep learning combined with multi-scale convolutional layers for extracting features from API sequences for malware detection was suggested in [14]. A solution for malware detection based on API subsequences of various lengths was proposed in [15].

Automated Windows malware detection via multi-scale feature fusion-based deep learning on API call sequences [16]

In [17], the concept of zero-shot learning for malware detection via API call sequences was exploited which enabled detection of new malware families without re-training the model.

The impact of convolutional neural networks and FastText embeddings on the task of text classification was considered in [18] where it was demonstrated that the model showed good results on benchmarking datasets.

The hybrid LSTM-CNN and CNN-LSTM models combined with GloVe embeddings were applied to multi-class sentiment classification of gender-based violence on Twitter in [19].

In [20], a hybrid approach consisting of a CNN and LSTM model for analyzing and detecting malware, which used convolutional and recurrent layers to learn spatial and sequential behavioral features from malware datasets, has been proposed. This model showed better detection results than other deep learning models that were used individually.

Different from the above papers, our approach makes use of three CNN designs – Single Kernel CNN-LSTM (with kernel size of 3), Multi-Kernels CNN (with kernel sizes of 3, 5, 7) and 5 Scale Multi-Kernels CNN (with kernel sizes of 3, 5, 7, 9, 11), for the purpose of ransomware detection. Moreover, in our study we propose Early Detection Analysis where the performances of different models are evaluated based on the increasing number of API calls observation windows.

III. METHODOLOGY

A. Dataset

In this study, a dataset was used which includes API call sequences generated from executing malware samples in the sandbox environment. In the dataset, there are 100 API call tokens per sample in numeric format labeled as either malware (1) or benign (0). There are 42,797 samples of malware and 1,079 samples of benign in the original dataset leading to an imbalanced dataset. To solve this problem, oversampling is performed on the benign class samples until the number of samples in each class equals 42,797, thereby generating a balanced dataset of 85,594 samples.

B. Data Preprocessing

Each sample in the dataset consists of 100 sequence of API calls in numeric form. This data was used as input to the embedding layer of the deep learning model without any further tokenization because numeric token was already available. Feature matrix X of size (85,594 × 100) and target vector y were created after removing the hash identifier and target columns respectively.

C. Proposed Models

The model that we propose is a 5-scale Multi-kernel CNN. The sequence of 100 API calls tokens is fed to the Embedding layer which converts each token to a 64D dense vector. Five parallel 1D convolutions of kernel sizes 3, 5, 7, 9 and 11 respectively are performed on the output of the embedding layer in parallel. GlobalMaxPooling operation follows each branch to get the most significant features. The outputs of all five branches are concatenated together to form a 320D feature vector. This vector is then processed by the Dense layer of size 128 with ReLU activation. It goes through Dropout of 0.5 before going through the output sigmoid layer. The number of trainable parameters of this network is 204,545.

D. Baseline Model

CNN-LSTM network architecture: CNN with kernel size 3 followed by Bi-directional LSTM (with 128 hidden units) followed by Dense followed by Sigmoid Three Scale CNN network architecture: Three convolutional layers with kernel sizes 3, 5, 7 followed by Global max pooling followed by Concatenation followed by Dense followed by Sigmoid.

E. Early Detection

To evaluate how early ransomware can be detected, API call tokens beyond position w were masked to zero for w 10, 20, 30, 50, 75, 100. Model accuracy measured at each window for CNN-LSTM and 5-Scale CNN. Detection curve generated showing relationship between observation length and detection performance.

F. Experimental Set-up

All experiments have been done in Google Colaboratory using Python 3 programming language along with TensorFlow 2.x and Keras. For the experiments, we have trained the model for 10 epochs with a batch size of 32 using Adam optimizer and binary cross-entropy as loss function. The validation split for the experiments is 15 percent. The dataset has been uploaded from Google Drive and all models have been saved in Keras format.

IV. Problem Statement

Given a stream of system-level behavioral events $\mathcal{E} = \{e_1, e_2, \ldots, e_T\}$ generated by a running process P on a host machine, the objective is to classify P as either *ransomware* or *benign* within the earliest possible window τ such that τ precedes the onset of mass file encryption.

Formally, let $\mathbf{x}^{(t)} \in \mathbb{R}^d$ be a feature vector encoding behavioral events at timestep t, and let $y \in \{0, 1\}$ be the binary label (benign = 0, ransomware = 1). The model f_θ must learn:

$$\hat{y} = f_\theta\left(\mathbf{x}^{(1)}, \mathbf{x}^{(2)}, \ldots, \mathbf{x}^{(\tau)}\right) \tag{1}$$

subject to the constraint that $\tau < t_{\text{encrypt}}$, where t_{encrypt} is the timestep at which the first file encryption write occurs.

The key challenges are:

1) **Temporal early detection:** The model must trigger before substantial damage—ideally within the first 5–10 seconds of process execution.
2) **Low false positive rate:** Legitimate applications (backup tools, archivers, file converters) exhibit superficially similar I/O patterns; the model must distinguish them.
3) **Generalization across ransomware families:** The model should detect novel variants not seen during training by learning family-agnostic behavioral signatures.
4) **Real-time inference latency:** Detection must complete within a latency budget compatible with endpoint deployment (<100 ms per inference window).

V. Results and Discussion

A. Classification Results Three deep neural networks were tested on a balanced test data set consisting of 17,119 examples. The CNN-LSTM network gave a classification accuracy of 99.30 percent with 89 false classifications. The 3-Scale Multi-Kernel CNN outperformed the former with an accuracy of 99.53 percent and 76 false classifications. The new 5-Scale Multi-Kernel CNN gave the highest accuracy of 99.59 percent with 60 false classifications. Confusion matrix for all the three deep learning models is presented in Fig. 1. It is evident that increasing the number of convolutional kernels results in improvement in the performance of the model due to multi-scale capture of API calls.

B. Early detection Analysis The Early Detection Analysis was carried out to determine how soon each model can able to detect ransomware from when process execution commences. From Fig. 2 below, it is clear that the 5-Scale Multi-Kernel CNN model achieves an accuracy level of 77.8 percent only 20 API calls as opposed to the CNN-LSTM which only has an accuracy level of 55.5 percent. This clearly shows that multi-scale feature extraction is effective in ensuring the ability to detect early. Both models achieve an accuracy level of over 99 percent by 100 API calls.

C. DISCUSSION The findings of the current study coincide with recent research. It has already been shown that the high efficiency of the methods based on the use of CNN and LSTM in malware detection [2], [20]. However, what sets the current study apart from the previous one is the use of multi-scale parallel CNN for ransomware detection alongside the analysis of the early detection. Thus, there emerges a possibility to measure the required time of detection, which was not taken into consideration earlier. The transition of CNN-LSTM to 3-scale and 5-scale CNN proves that kernel diversity leads to behavior diversity.

TABLE I
Comparison of Model Performance

Model	Accuracy	F1-Score	Errors
CNN-LSTM	99.30%	0.99	89
3-Scale CNN	99.53%	1.00	76
5-Scale CNN	99.59%	1.00	60

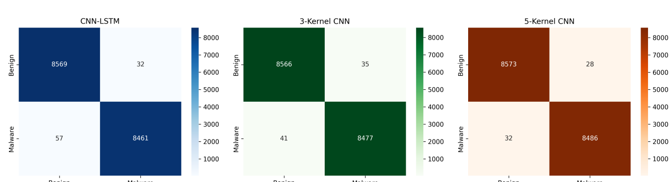

Fig. 1. Confusion Matrices - CNN-LSTM, 3-Kernel CNN, 5-Kernel CNN

TABLE II
EARLY DETECTION ACCURACY AT DIFFERENT WINDOWS

API Calls	CNN-LSTM	5-Scale CNN
10	54%	54%
20	55.5%	77.8%
30	58%	77%
50	65%	74%
75	86.7%	79.2%
100	99.48%	99.65%

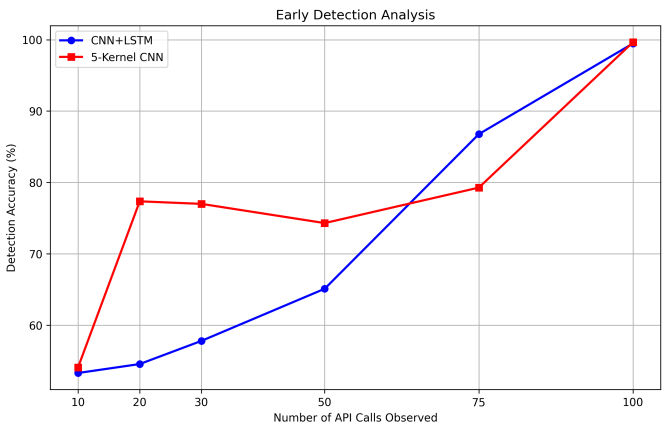

Fig. 2. Early Detection Analysis

VI. CONCLUSION

This paper proposed a deep learning based framework for detecting ransomware before the file encryption phase begins by analyzing Windows API call sequences. Three models were implemented and compared — a CNN-LSTM baseline, a 3-Scale Multi-Kernel CNN, and the proposed 5-Scale Multi-Kernel CNN with parallel convolutional branches of kernel sizes 3, 5, 7, 9, and 11. All models were trained and evaluated on a balanced dataset of 85,594 API call sequence samples. The proposed 5-Scale CNN achieved the best classification accuracy of 99.59 percent with only 60 misclassifications on the test set of 17,119 samples. An Early Detection Analysis was introduced , demonstrating that the proposed model detects ransomware with 77.8 percent accuracy after observing just 20 API calls — well before file encryption begins. Results confirm that increasing the number of parallel convolutional kernels progressively improves both final accuracy and early detection capability, making multi-scale CNN architectures a strong choice for proactive ransomware detection.The proposed framework shows that multi-scale deep learning for behavioral analysis is a good approach to proactive endpoint security for ransomware.

VII. FUTURE WORK

Several directions have been identified for future work. Firstly, the effectiveness of the proposed framework would be tested on different datasets which would include raw behavior logs generated from sandbox environments. Secondly, the framework would be used for classification at the level of families in order to detect specific variants of ransomware rather than just binary classification. Thirdly, an attention-based approach would be used in order to increase the explainability of the system by pointing out important API calls within the sequence. Fourthly, the framework would be optimized for deployment in real time on endpoints. Fifthly, the proposed model's robustness against evasion techniques would be studied.Furthermore, the proposed framework can also be extended to detecting ransomware in mobile and IoT environments where there is an increasing threat of ransomware attacks.

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